



2009

**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Chemistry

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A data sheet and a Periodic Table are provided at the back of this paper
- Write your Centre Number and Student Number at the top of pages 9, 11, 13, 15 and 17

Total marks – 100

Section I Pages 2–20

75 marks

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–26
- Allow about 1 hour and 45 minutes for this part

Section II Pages 21–33

25 marks

- Attempt ONE question from Questions 27–31
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 15 marks

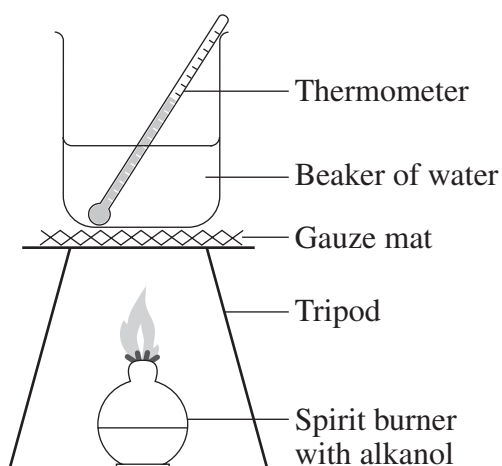
Attempt Questions 1–15

Allow about 30 minutes for this part

Use the multiple-choice answer sheet for Questions 1–15.

- 1 Which of the following is an important factor in predicting the nuclear stability of an isotope?
- (A) Atomic radius
 - (B) Nuclear radius
 - (C) The ratio of neutrons to protons
 - (D) The ratio of electrons to protons
- 2 Unpolluted rain water in New South Wales is slightly acidic.
- Which substance is the major contributor to this acidity?
- (A) Ozone
 - (B) Sulfur dioxide
 - (C) Carbon dioxide
 - (D) Nitrogen dioxide
- 3 Which of the following groups contains ONLY acidic substances?
- (A) Antacid tablets, baking soda, laundry detergents
 - (B) Blood, oven cleaner, seawater
 - (C) Milk, tea, drain cleaner
 - (D) Vinegar, wine, aspirin

- 4 What flame colour is produced by barium ions in a flame test?
- (A) Red
 - (B) Blue
 - (C) Green
 - (D) Orange
- 5 The apparatus shown is used in a first-hand investigation to determine and compare the heat of combustion of three different liquid alkanols.



Which is the independent variable?

- (A) Type of alkanol used
- (B) Amount of water used
- (C) Amount of alkanol used
- (D) Temperature change in the water

- 6 Bromine, Br_2 , dissolves in unsaturated hydrocarbons and reacts immediately.

Which of the following is the best description of this process?

- (A) Bromine is polar and reacts by adding bromine atoms across the double bond.
- (B) Bromine is polar and reacts by substituting hydrogen atoms with bromine atoms.
- (C) Bromine is non-polar and reacts by substituting hydrogen atoms with bromine atoms.
- (D) Bromine is non-polar and reacts by adding bromine atoms across the double bond.

- 7 What is the conjugate base of HSO_4^- ?

- (A) SO_3^{2-}
- (B) SO_4^{2-}
- (C) H_2SO_4
- (D) HSO_3^-

- 8 Three separate colourless solutions each contain one cation, Na^+ , Pb^{2+} or Ca^{2+} .

Which of the following would be an appropriate reagent to unambiguously identify the solution containing Pb^{2+} ?

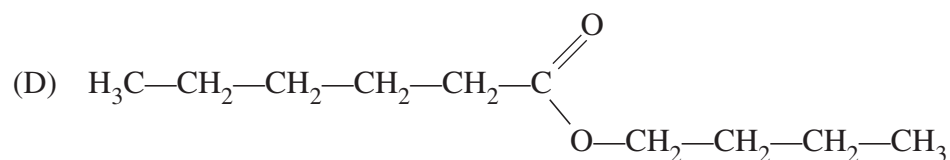
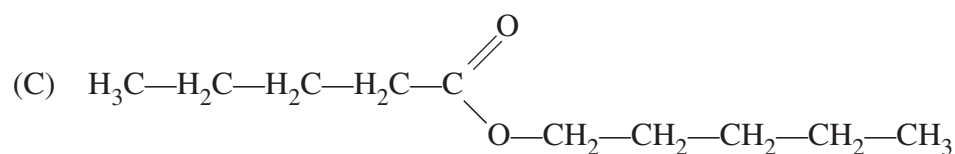
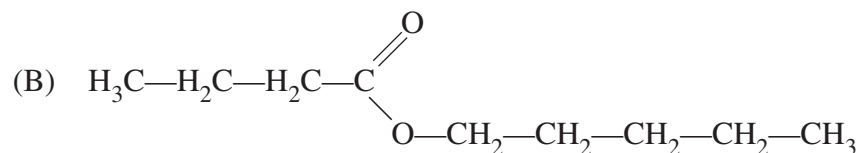
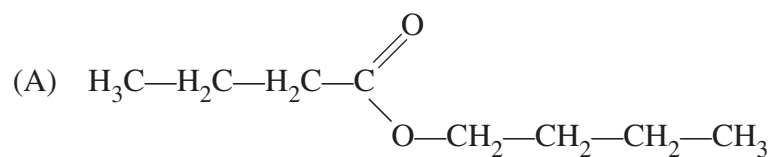
- (A) KI
- (B) K_2CO_3
- (C) K_3PO_4
- (D) AgNO_3

- 9 One test used for random breath testing in NSW involved crystals of potassium dichromate reacting with ethanol. In this reaction the orange dichromate ion, $\text{Cr}_2\text{O}_7^{2-}$, changes to the green chromium ion, Cr^{3+} .

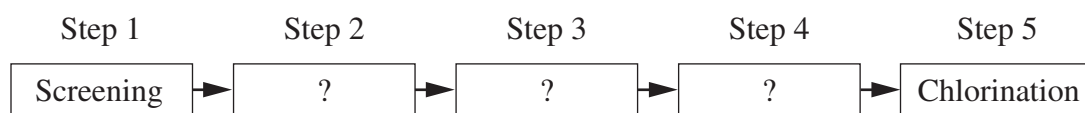
Which statement is true for this reaction?

- (A) Chromium has lost electrons and reached a lower oxidation state.
- (B) Chromium has lost electrons and reached a higher oxidation state.
- (C) Chromium has gained electrons and reached a lower oxidation state.
- (D) Chromium has gained electrons and reached a higher oxidation state.

- 10 Which of the following is the main organic product resulting from the reaction of butanoic acid and pentanol?



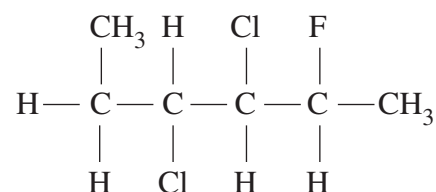
- 11 The following process is used to purify water for drinking.



Which sequence represents the correct order of Steps 2, 3 and 4?

	<i>Step 2</i>	<i>Step 3</i>	<i>Step 4</i>
(A)	Flocculation	pH adjustment	Settling
(B)	pH adjustment	Flocculation	Settling
(C)	pH adjustment	Settling	Flocculation
(D)	Flocculation	Settling	pH adjustment

- 12 What is the IUPAC name of the following compound?



- (A) 3,4-dichloro-2-fluorohexane
(B) 3,4-dichloro-5-fluorohexane
(C) 2-fluoro-3,4-dichlorohexane
(D) 5-fluoro-3,4-dichlorohexane
- 13 In a fermentation experiment 6.50 g of glucose was completely converted to ethanol and carbon dioxide.

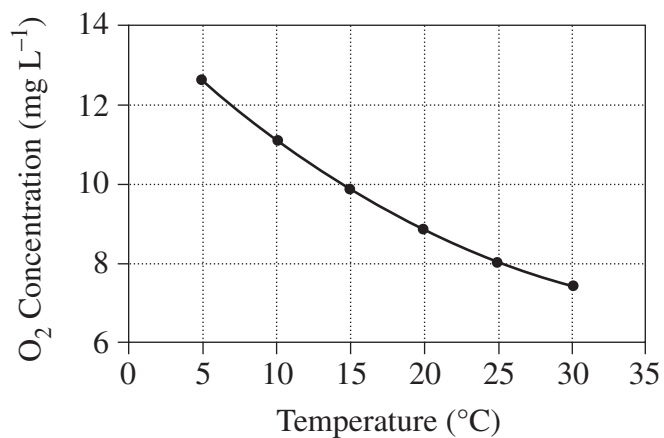
What is the mass of carbon dioxide produced?

- (A) 1.59 g
(B) 3.18 g
(C) 9.53 g
(D) 13.0 g
- 14 Citric acid, the predominant acid in lemon juice, is a triprotic acid. A student titrated 25.0 mL samples of lemon juice with 0.550 mol L⁻¹ NaOH. The mean titration volume was 29.50 mL. The molar mass of citric acid is 192.12 g mol⁻¹.

What was the concentration of citric acid in the lemon juice?

- (A) 1.04 g L⁻¹
(B) 41.6 g L⁻¹
(C) 125 g L⁻¹
(D) 374 g L⁻¹

- 15 The graph shows the maximum dissolved oxygen concentration in water as a function of temperature at normal atmospheric pressure.



What is the volume of O₂ that can dissolve in 10.0 L of water at 25°C and normal atmospheric pressure?

- (A) 62.0 mL
- (B) 63.5 mL
- (C) 80.0 mL
- (D) 124 mL

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Centre Number

Section I (continued)

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Part B – 60 marks

Student Number

Attempt Questions 16–26

Allow about 1 hour and 45 minutes for this part

Answer the questions in the spaces provided.

Show all relevant working in questions involving calculations.

Question 16 (3 marks)

Describe how to prepare an ester in the school laboratory. Include a specific safety precaution in your answer.

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Question 17 (4 marks)

Water and ethanol are both used as solvents.

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Explain the differences and similarities in their solvent behaviour in terms of their molecular structures. Include a diagram in your answer.

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Centre Number

Section I – Part B (continued)

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Student Number

Question 18 (5 marks)

There has been an increase in the concentration of the oxides of nitrogen in the atmosphere as a result of combustion.

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Assess both the evidence to support this statement and the need to monitor these oxides.

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Question 19 (6 marks)

Outline the chemical and physical processes involved in the production of ONE of the following from a natural raw material:

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- a polyethylene bottle
- a polyvinyl chloride pipe
- a polystyrene cup.

Include relevant chemical equations in your answer.

[illegible]

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Centre Number

Section I – Part B (continued)

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Student Number

Question 20 (4 marks)

- (a) Calculate the mass of ethanol that must be burnt to increase the temperature of 210 g of water by 65°C, if exactly half of the heat released by this combustion is lost to the surroundings. 3

The heat of combustion of ethanol is 1367 kJ mol⁻¹.

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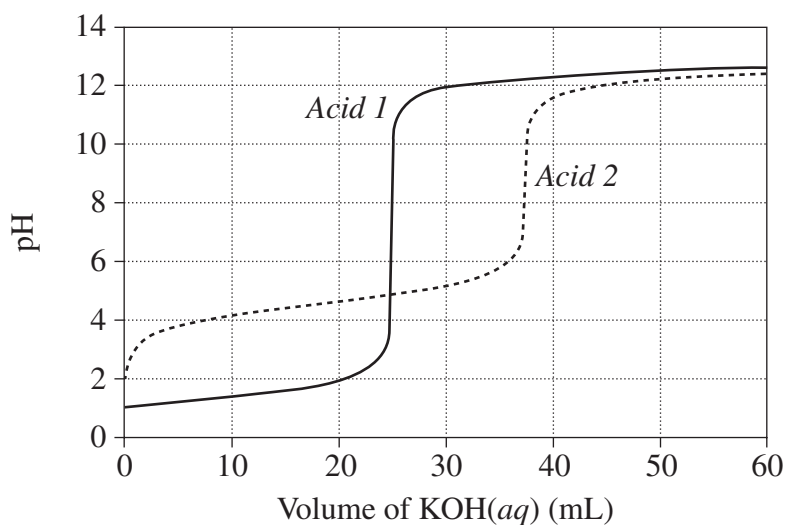
- (b) What are TWO ways to limit heat loss from the apparatus when performing a first-hand investigation to determine and compare heat of combustion of different liquid alkanols? 1

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Question 21 (6 marks)

The graph shows changes in pH for the titrations of equal volumes of solutions of two monoprotic acids, *Acid 1* and *Acid 2*.



- (a) Explain the differences between *Acid 1* and *Acid 2* in terms of their relative strengths and concentrations. **3**

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- (b) Name the salt produced by the reaction of an acid of the same type as *Acid 2* with KOH(aq) . **1**

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- (c) Calculate the concentration of hydrogen ions when 20 mL of KOH(aq) has been added to *Acid 1*. **1**

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- (d) Why would phenolphthalein be a suitable indicator for both titrations? **1**

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Centre Number

Section I – Part B (continued)

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Student Number

Question 22 (7 marks)

The nitrogen content of bread was determined using the following procedure:

- A sample of bread weighing 2.80 g was analysed.
- The nitrogen in the sample was converted into ammonia.
- The ammonia was collected in 50.0 mL of 0.125 mol L⁻¹ hydrochloric acid. All of the ammonia was neutralised, leaving an excess of hydrochloric acid.
- The excess hydrochloric acid was titrated with 23.30 mL of 0.116 mol L⁻¹ sodium hydroxide solution.

- (a) Write balanced equations for the TWO reactions involving hydrochloric acid. 2

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- (b) Calculate the moles of excess hydrochloric acid. 1

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- (c) Calculate the moles of ammonia. 2

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- (d) Calculate the percentage by mass of nitrogen in the bread. 2

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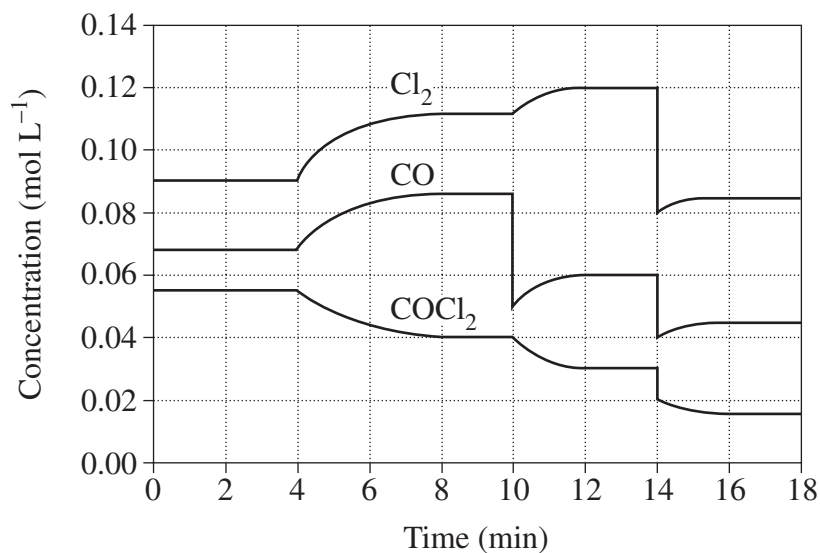
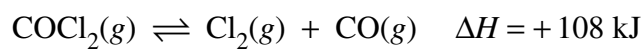
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Question 23 (6 marks)

The graph shows the variation in concentration of reactant and products as a function of time for the following system.

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Identify and explain each of the changes in conditions that have shaped the curves during the time the system was observed.

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Centre Number

Section I – Part B (continued)

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Student Number

Question 24 (5 marks)

Describe the principle of atomic absorption spectroscopy and its application in environmental monitoring. Include a diagram in your answer.

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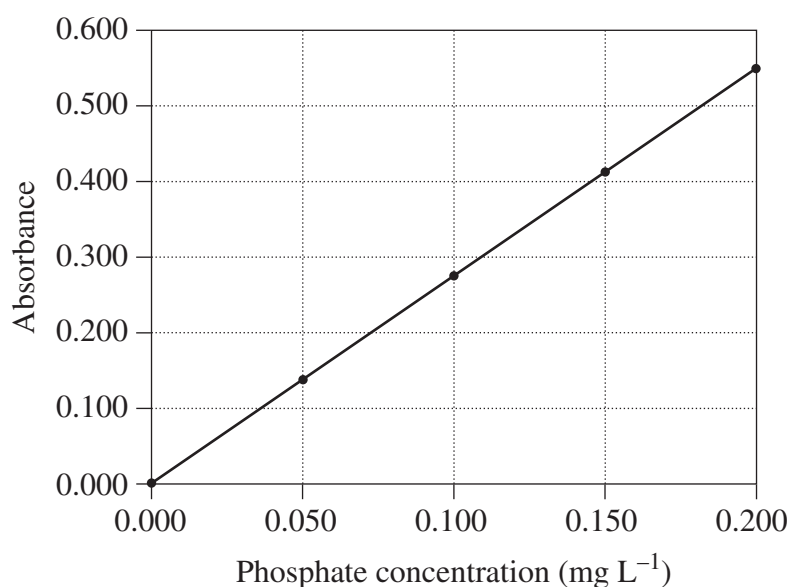
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Question 25 (7 marks)

An analytical chemist determined the phosphate concentration of water samples from three local streams.

- (a) Using the absorbance values in the table and graph, determine the mean absorbance and mean phosphate concentration for each stream and complete the table. 2

<i>Stream</i>	<i>Absorbances measured</i>	<i>Mean absorbance</i>	<i>Mean phosphate concentration (mg L⁻¹)</i>
1	0.090, 0.092, 0.088		
2	0.513, 0.511, 0.514		
3	0.234, 0.237, 0.234		



Question 25 continues on page 19

Question 25 (continued)

- (b) The recommended maximum level of phosphate in streams is 0.100 mg L^{-1} .

With reference to the recommended maximum level of phosphate for stream water, explain why there are differences between the three streams.

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- (c) Why is phosphate concentration a water quality issue?

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End of Question 25

Question 26 (7 marks)

An electrochemical cell is constructed using two half cells. One half cell consists of an inert platinum electrode and a solution of Fe^{2+} and Fe^{3+} . The other half cell consists of a lead electrode and a solution of Pb^{2+} .

Current will flow from one electrode to the other electrode when the cell is completed using a voltmeter and a salt bridge.

- (a) Write relevant half equations and a balanced net ionic equation for the overall cell reaction. **2**

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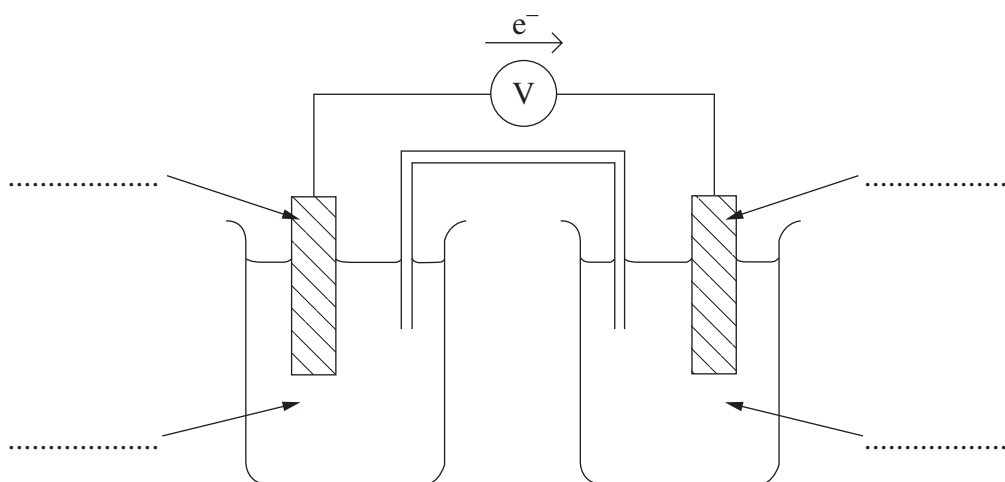
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- (b) Calculate the standard cell potential ($E^\ominus$). **1**

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- (c) Identify the anode, cathode, metals and ions by labelling the following diagram. **3**



- (d) Identify an appropriate electrolyte to use in the salt bridge. **1**

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Chemistry

Section II

25 marks

Attempt ONE question from Questions 27–31

Allow about 45 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

Show all relevant working in questions involving calculations.

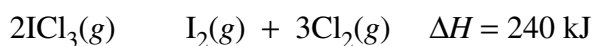
	Pages
Question 27 Industrial Chemistry	22–23
Question 28 Shipwrecks, Corrosion and Conservation	24–25
Question 29 The Biochemistry of Movement	26–27
Question 30 The Chemistry of Art	28–30
Question 31 Forensic Chemistry	31–33

Question 27 — Industrial Chemistry (25 marks)

- (a) Sulfuric acid is one of the world's most significant industrial chemicals because of the variety and importance of its uses.

- (i) Identify the major use of sulfuric acid. **1**
- (ii) Outline the industrial process for the manufacture of sulfuric acid from its raw materials. **3**
- (iii) Account for the safety precautions associated with the industrial transport of sulfuric acid. **2**

- (b) At a particular temperature, iodine trichloride dissociates into iodine gas and chlorine gas according to the following equation:



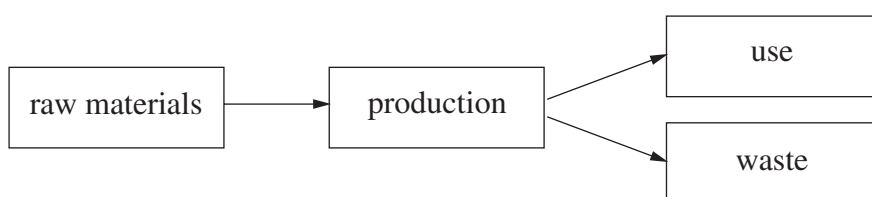
Initially 0.35 mol of $\text{ICl}_3(\text{g})$ was introduced into a 1.0 L container and allowed to come to equilibrium. At equilibrium there was 0.45 mol L^{-1} of $\text{Cl}_2(\text{g})$.

- (i) Write the equilibrium constant expression for this reaction. **1**
- (ii) Calculate the value of K at this temperature. **3**
- (iii) What are TWO consequences of increasing the temperature of the mixture at equilibrium? **2**

Question 27 continues on page 23

Question 27 (continued)

- (c) Account for a use of an emulsion in terms of its properties. **2**
- (d) (i) Explain the cleaning action of soap in terms of its molecular structure. **2**
- (ii) Soap is one product of saponification. Name the other product and draw its structural formula. **2**
- (e) The flowchart summarises the fundamental criteria that must be considered in order to find a suitable location for an industrial plant. **7**

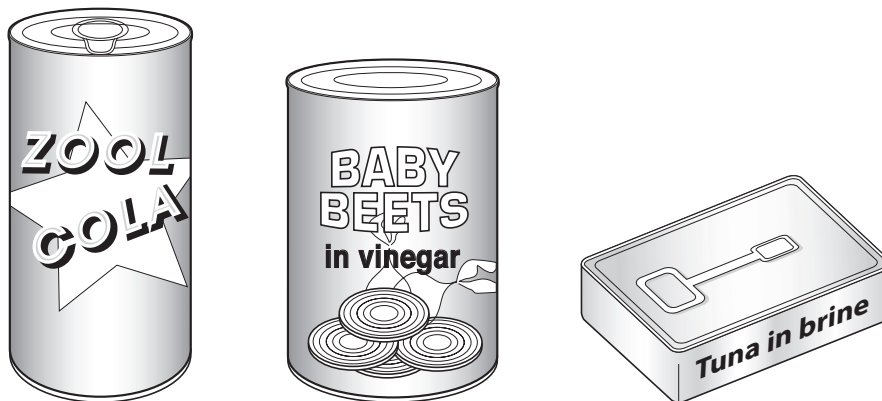


With reference to the flowchart, explain the significance of each criterion to determine a suitable location for an industrial plant to manufacture sodium carbonate.

End of Question 27

Question 28 — Shipwrecks, Corrosion and Conservation (25 marks)

- (a) Tins and cans are used to store a range of foods, beverages and general household chemicals.



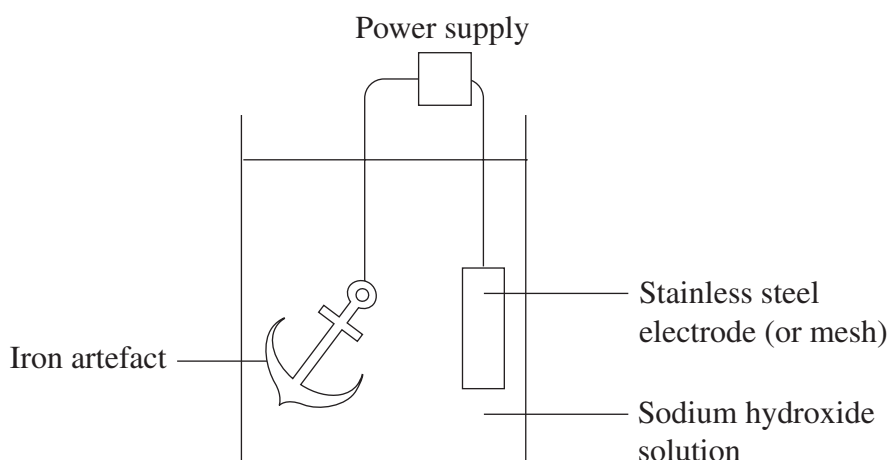
- (i) Identify the passivating metal used in drink cans in Australia. **1**
- (ii) Compare the use and effectiveness in preventing corrosion of different protective coatings on containers, such as those shown. **3**
- (b) Galvanising is used outside the home on water tanks and fencing. Explain the protection provided to iron by galvanising. **2**
- (c) The roof guttering on a garage has rusted through.
- (i) Explain, using chemical equations, the cause of this problem. **2**
- (ii) The owners have decided to replace the guttering. They have steel screws and a choice of aluminium or copper guttering. **4**

Justify the course of action the owners should take, based on chemical principles.

Question 28 continues on page 25

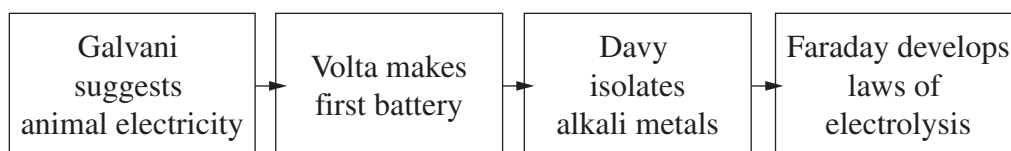
Question 28 (continued)

- (d) (i) The diagram shows an electrolytic cell used to conserve an iron artefact recovered from the wreck of a ship that sank in the early 1800s. **3**



Explain how this process is used to conserve this iron artefact.

- (ii) Describe, with the aid of a diagram, how a nickel spoon could be silver-plated using equipment available in a school laboratory. **3**
- (e) The flowchart summarises the historical development of the understanding of electron transfer reactions. **7**



With reference to the flowchart, describe how the work of the scientists led to a better understanding of electron transfer reactions.

End of Question 28

Question 29 — The Biochemistry of Movement (25 marks)

- (a) (i) Identify a factor that will cause denaturation of a protein molecule. **1**
- (ii) Explain denaturation of a protein in terms of its structure and bonding. **2**
- (b) All enzymes are proteins. Polyphenoloxidase (PPO) is an enzyme that catalyses the browning of fruit and vegetables. This catalytic action can be studied in the laboratory by monitoring the oxidation of a phenol to a quinone, a red compound. The intensity of the red colour indicates the extent of the reaction. A student studied the enzyme activity of PPO in relation to a number of biological factors, using quinone as the measure of activity. **3**

The following table summarises the results of one set of these experiments.

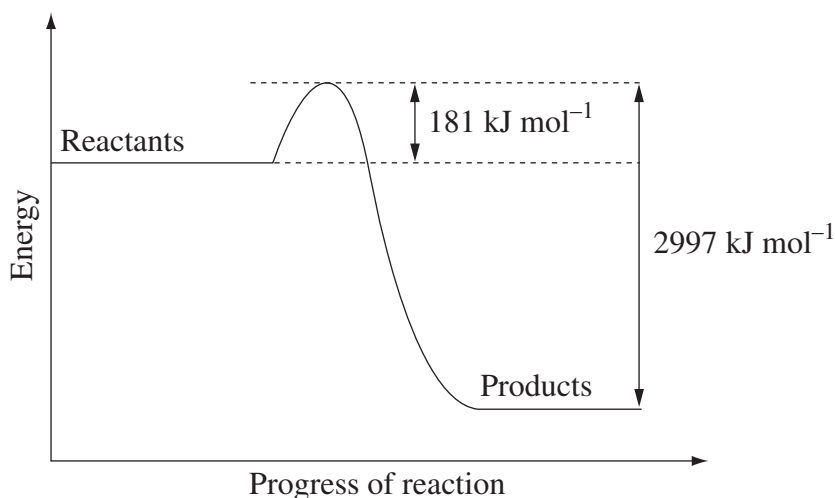
<i>pH at 37°C</i>	3.0	4.0	5.0	6.0	7.0	8.0
<i>Colour</i>	colourless	light red	medium red	dark red	dark red	colourless

Account for this set of experimental results in terms of the characteristics of reactions involving enzymes.

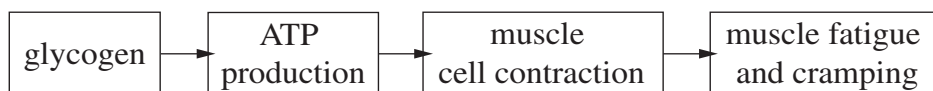
Question 29 continues on page 27

Question 29 (continued)

- (c) The following energy diagram shows the energy change associated with the combustion of glucose in kJ mol^{-1} during cellular respiration.



- (i) Write a balanced chemical equation for the combustion of glucose. Include the energy released per mole of glucose. **2**
- (ii) Calculate the energy released per gram of glucose. **2**
- (iii) An average cyclist uses 80 kJ for every km travelled. If the glucose content of an energy bar is 36.7 g, how far can an average cyclist travel on only one energy bar? **2**
- (d) Analysing the structure of the glycerol molecule helps explain many of its properties. **3**
- (i) Explain the viscosity of glycerol and its solubility in water. **3**
- (ii) Explain the solubility of both fatty acids and triacylglycerols in water in terms of their molecular structures. **3**
- (e) The flowchart summarises a particular biochemical pathway related to muscle cell respiration. **7**



With reference to the flowchart, describe how a knowledge of the biochemical reactions involved in muscle cell respiration has led to more informed training programs. Include relevant chemical equations in your answer.

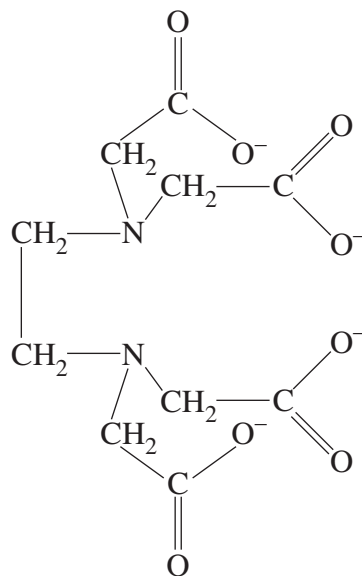
End of Question 29

Question 30 — The Chemistry of Art (25 marks)

- (a) (i) Name a monodentate ligand.

1

The ethylenediaminetetraacetate ion (EDTA) is a polydentate ligand. It is sometimes used to treat people suffering from heavy metal poisoning.



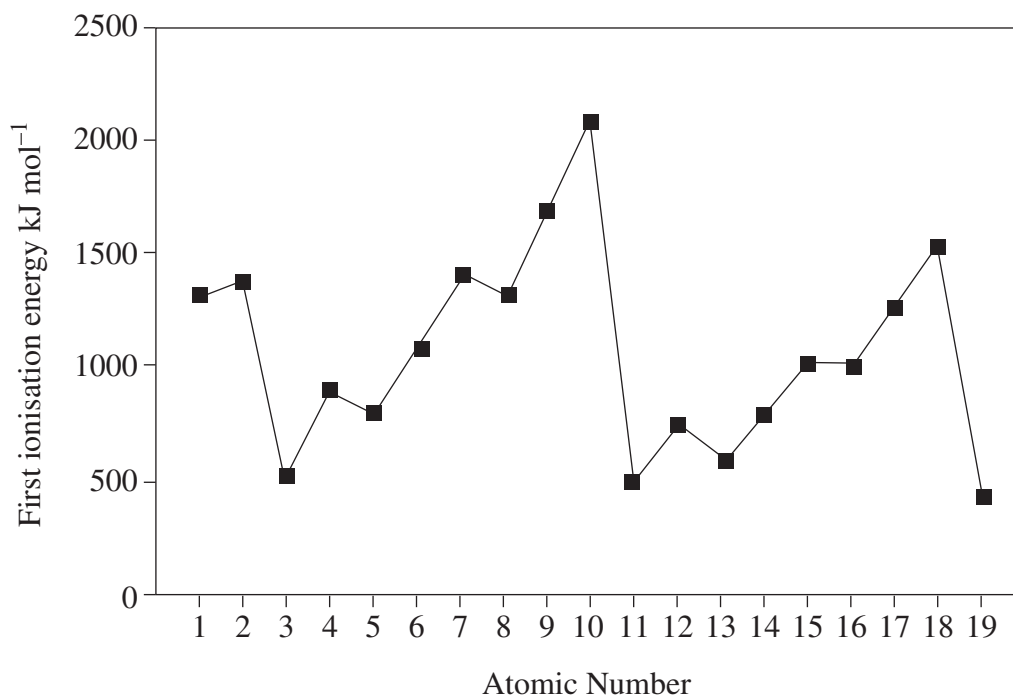
EDTA

- (ii) With reference to EDTA, explain what is meant by the term *polydentate ligand*. **2**
- (iii) Using copper as the ion and EDTA as the ligand, explain how bonding occurs between a metal ion and a ligand. **3**

Question 30 continues on page 29

Question 30 (continued)

- (b) (i) What is the maximum number of electrons that a p-orbital can hold? **1**
- (ii) Write the full electron configurations for a Ca atom in the ground state, an excited Ca atom and a Ca^+ ion. **2**
- (c) The graph shows the first ionisation energy of some elements. **3**



Account for the trends in the graph in terms of the electron configuration of the elements.

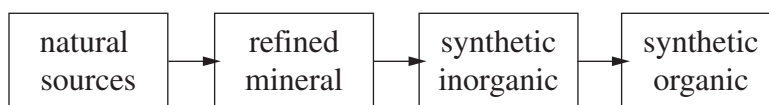
- (d) (i) Gustav Kirchhoff formulated a law that states that ‘a hot gas produces light with spectral lines at discrete wavelengths’. **3**
- Explain the chemical principle that underpins this law.
- (ii) Describe how a first-hand investigation in the school laboratory could be undertaken to determine the presence of a particular metal in a water sample using a flame test. Give a reason why this method may not work. **3**

Question 30 continues on page 30

Question 30 (continued)

- (e) The flowchart summarises the development of pigments over time.

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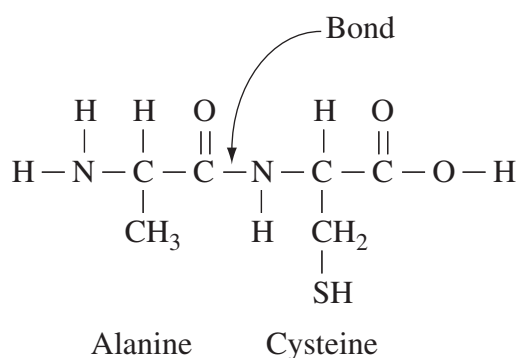
Historically many pigments have contained elements with atomic numbers in the range of 22 to 30.

With reference to the flowchart, describe the use of pigments containing these elements and the origin of pigment colour at the atomic level.

End of Question 30

Question 31 — Forensic Chemistry (25 marks)

- (a) (i) This structure represents a biological molecule.



Name the specific type of covalent bond indicated.

1

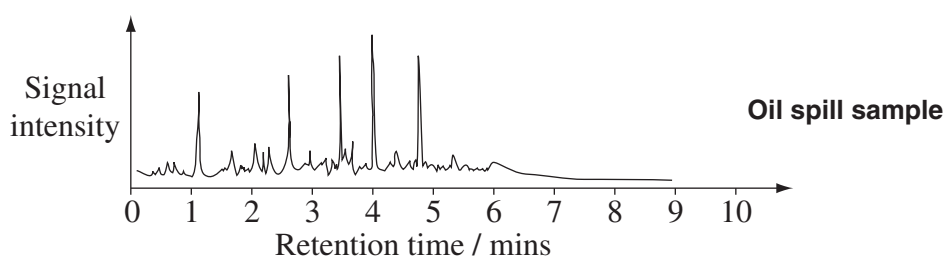
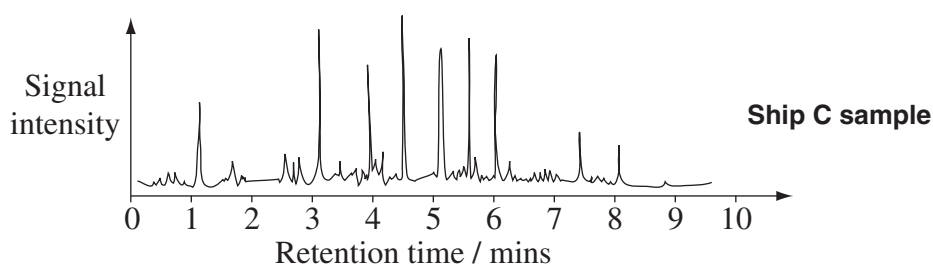
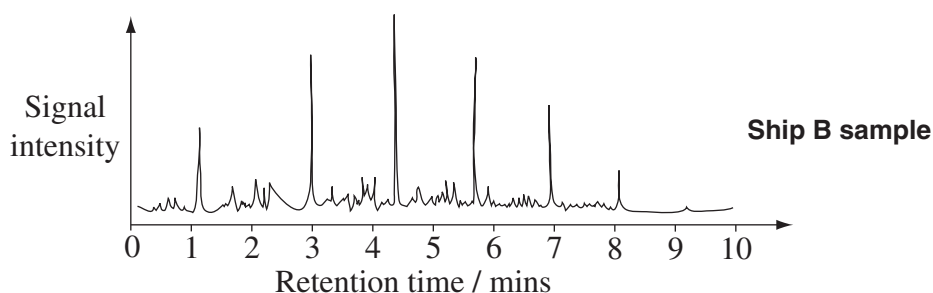
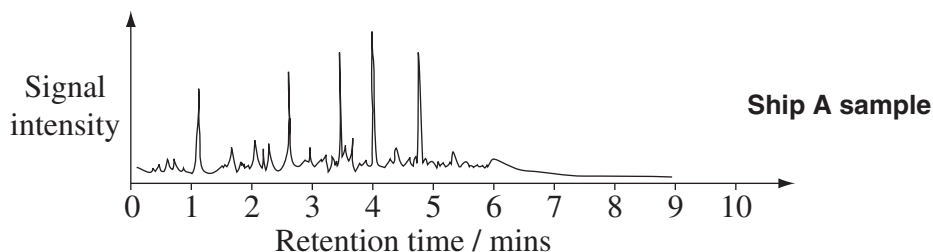
- (ii) Distinguish between the primary, secondary and tertiary structures of proteins. **3**
- (iii) How could electrophoresis be used to identify the origins of proteins in an investigation of adulterated food? **2**

Question 31 continues on page 32

Question 31 (continued)

- (b) An oil spill occurred at a shipping port. To determine which ship was responsible, a sample of the oil was collected and analysed by gas chromatography. Samples of bunker oil were collected from three ships in port at the time and analysed by gas chromatography.

The chromatograms of the samples collected are shown.

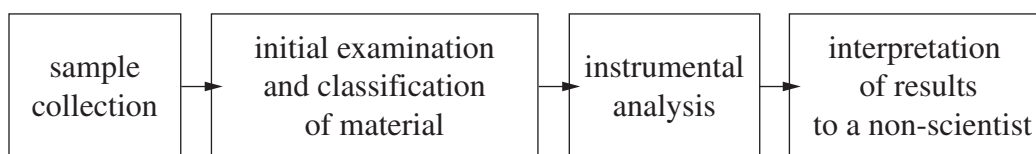


- (i) By comparing the chromatograms identify whether the oil spill originated from one of these vessels. **1**
- (ii) Explain what the peaks in the chromatograms represent. **2**
- (c) Describe the features of instrumental chromatography (GLC or HPLC) that allow the analysis of small samples. **3**

Question 31 continues on page 33

Question 31 (continued)

- (d) (i) Describe how a first-hand investigation in the school laboratory could be used to separate and identify the components of a mixed food dye. **3**
- (ii) Describe the chemical tests that could be used to distinguish between alkenes, alkanols and alkanolic acids. **3**
- (e) The flowchart summarises the steps involved in sample processing and presentation of results for a forensic investigation. **7**



You are the forensic chemist investigating a crime scene and collecting samples. With reference to the flowchart, describe how you would process the sample and present the results. Illustrate your answer with relevant tests and appropriate analyses.

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DATA SHEET

Avogadro constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0°C (273.15 K)	22.71 L
at 25°C (298.15 K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+]$$

$$\Delta H = -m C \Delta T$$

Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008 Hydrogen		KEY																2 He 4.003 Helium																	
3 Li 6.941 Lithium		4 Be 9.012 Beryllium		Atomic Number		79 Au 197.0 Gold		Symbol of element		5 B 10.81 Boron		6 C 12.01 Carbon		7 N 14.01 Nitrogen		8 O 16.00 Oxygen		9 F 19.00 Fluorine		10 Ne 20.18 Neon															
		12 Mg 24.31 Magnesium		Atomic Weight		Name of element				13 Al 26.98 Aluminium		14 Si 28.09 Silicon		15 P 30.97 Phosphorus		16 S 32.07 Sulfur		17 Cl 35.45 Chlorine		18 Ar 39.95 Argon															
19 K 39.10 Potassium		20 Ca 40.08 Calcium		21 Sc 44.96 Scandium		22 Ti 47.87 Titanium		23 V 50.94 Vanadium		24 Cr 52.00 Chromium		25 Mn 54.94 Manganese		26 Fe 55.85 Iron		27 Co 58.93 Cobalt		28 Ni 58.69 Nickel		29 Cu 63.55 Copper		30 Zn 65.41 Zinc		31 Ga 69.72 Gallium		32 Ge 72.64 Germanium		33 As 74.92 Arsenic		34 Se 78.96 Selenium		35 Br 79.90 Bromine		36 Kr 83.80 Krypton	
37 Rb 85.47 Rubidium		38 Sr 87.62 Strontium		39 Y 88.91 Yttrium		40 Zr 91.22 Zirconium		41 Nb 92.91 Niobium		42 Mo 95.94 Molybdenum		43 Tc [97.91] Technetium		44 Ru 101.1 Ruthenium		45 Rh 102.9 Rhodium		46 Pd 106.4 Palladium		47 Ag 107.9 Silver		48 Cd 112.4 Cadmium		49 In 114.8 Indium		50 Sn 118.7 Tin		51 Sb 121.8 Antimony		52 Te 127.6 Tellurium		53 I 126.9 Iodine		54 Xe 131.3 Xenon	
55 Cs 132.9 Caesium		56 Ba 137.3 Barium		57–71 Lanthanoids		72 Hf 178.5 Hafnium		73 Ta 180.9 Tantalum		74 W 183.8 Tungsten		75 Re 186.2 Rhenium		76 Os 190.2 Osmium		77 Ir 192.2 Iridium		78 Pt 195.1 Platinum		79 Au 197.0 Gold		80 Hg 200.6 Mercury		81 Tl 204.4 Thallium		82 Pb 207.2 Lead		83 Bi 209.0 Bismuth		84 Po [209.0] Polonium		85 At [210.0] Astatine		86 Rn [222.0] Radon	
87 Fr [223]		88 Ra [226]		89–103 Actinoids		104 Rf [261] Rutherfordium		105 Db [262]		106 Sg [266] Seaborgium		107 Bh [264] Bohrium		108 Hs [277] Hassium		109 Mt [268] Meitnerium		110 Ds [271] Darmstadtium		111 Rg [272] Roentgenium															

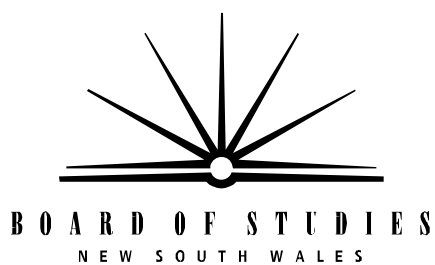
Lanthanoids

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm [145] Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.0 Ytterbium	71 Lu 175.0 Lutetium
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Actinoids

89 Ac [227] Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np [237] Neptunium	94 Pu [244] Plutonium	95 Am [243] Americium	96 Cm [247] Curium	97 Bk [247] Berkelium	98 Cf [251] Californium	99 Es [252] Einsteinium	100 Fm [257] Fermium	101 Md [258] Mendelevium	102 No [259] Nobelium	103 Lr [262] Lawrencium
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For elements that have no stable or long-lived nuclides, the mass number of the nuclide with the longest confirmed half-life is listed between square brackets.
The International Union of Pure and Applied Chemistry Periodic Table of the Elements (October 2005 version) is the principal source of data. Some data may have been modified.



2009 HSC Chemistry Marking Guidelines

Section I

Question	Correct Response
1	C
2	C
3	D
4	C
5	A
6	D
7	B
8	A
9	C
10	B
11	D
12	A
13	B
14	B
15	A

Section I, Part B

Question 16 (a)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
- Identifies appropriate name of reactants (alkanol+alkanoic acid) and catalyst (concentrated sulfuric acid) - Provides either the features of method, including reflux, or draws diagram - Identifies ONE relevant safety precaution	3
Gives any TWO of the above	2
- Identifies any TWO of alkanoic acid, alkanol and sulfuric acid OR - Identifies method is reflux(ing) OR - Identifies condenser is required OR - Identifies a relevant safety precaution	1

Question 17

Outcomes assessed: H2, H4, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Identifies water and ethanol are polar molecules that form hydrogen bonds which can dissolve polar substances - Identifies ethanol forms dispersion forces, therefore a good solvent for both polar and non-polar substances - Draws correct diagram showing hydrogen bonds for both molecules	4
TWO of the above	3
- Identifies both water and ethanol are polar molecules that form hydrogen bonds OR - States water dissolves polar substances, ethanol dissolves polar and non-polar OR - Draws correct diagram for both molecules	2
- Identifies water or ethanol as a polar molecule OR - Identifies ethanol as a polar and non-polar molecule OR - Draws correct diagram for one molecule	1

Question 18

Outcomes assessed: H4, H8, H6

MARKING GUIDELINES

Criteria	Marks
- Provides sources of evidence of increase and reason for monitoring - Provides sources of oxides of nitrogen - Provides balanced relevant equations - Describes TWO or more effects of oxide of nitrogen in the atmosphere - Provides a clear judgement based upon the above criteria	4–5
- Describes TWO or more effects of oxide of nitrogen in the atmosphere AND EITHER - Provides relevant equations OR - Provides sources of NO₂ leading to ONE effect OR - Provides evidence for the increase	2–3
- Identifies ONE oxide of nitrogen, eg NO₂ OR - Identifies ONE effect, eg photochemical smog OR - Identifies the industrial revolution and more cars have led to increased combustion and more nitrogen oxides OR - Identifies a piece of evidence to support increase OR - ONE relevant equation	1

Question 19

Outcomes assessed: H9, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Indicates all of the main chemical and physical processes from a raw material to a monomer to polymer to finished product from natural raw material - Demonstrates a thorough knowledge of the chemistry involved using relevant chemical equations - Presents in a logical and coherent manner	5–6
- Indicates some of the main chemical and physical processes involved in the production of the finished product from natural raw material - Demonstrates knowledge of some of the chemistry involved using chemical equations	3–4
- Indicates a feature in the manufacturing process of an identified product and provides an appropriate word equation OR - Indicates TWO features in the manufacturing process of identified product OR - Provides ONE correct formula and ONE feature of the manufacturing process OR - Provides a structural formula for a monomer and corresponding polymer	2
- Indicates a feature in the manufacturing process of an identified product OR - Provides an appropriate word equation OR - Provides an appropriate structural formula	1

Question 20 (a)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
Correctly calculates the mass of ethanol burnt	3
- Correctly calculates q - Correctly calculates the mass of ethanol WITHOUT consideration of heat loss not accounted for OR - Uses THREE correct steps with incorrect answer(s)	2
- Correctly calculates q with correct units OR - Incorrectly calculates mass and/or moles but demonstrates understanding of heat loss in calculation OR - Uses TWO correct steps with incorrect answers	1

Question 20 (b)

Outcomes assessed: H11

MARKING GUIDELINES

Criteria	Marks
States TWO appropriate ways to limit heat loss	1

Question 21 (a)

Outcomes assessed: H12, H14

MARKING GUIDELINES

Criteria	Marks
- Relates pH of equivalence point to the nature of each acid correctly - Relates concentration to amount of base used	3
- One of the above for both acids OR - States strong, less concentrated acid – Acid 1 - States weak, more concentrated acid – Acid 2	2
- Identifies strength of one acid correctly OR - Identifies relative concentration of one acid correctly OR - Identifies equivalence point of either acid correctly OR - States that Acid 2 uses more base than Acid 1	1

Question 21 (b)

Outcomes assessed: H8, H10

MARKING GUIDELINES

Criteria	Marks
Correctly identifies the alkaline salt by name or formula relative to part (a)	1

Question 21 (c)

Outcomes assessed: H12, H13

MARKING GUIDELINES

Criteria	Marks
Correctly calculates hydrogen ion concentration with correct units	1

Question 21 (d)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
Correctly relates colour change range for phenolphthalein to equivalence point or inflexion point on graph or vertical section of curve	1

Question 22 (a)

Outcomes assessed: H10, H13

MARKING GUIDELINES

Criteria	Marks
• Gives correctly balanced equations	2
• Gives one correctly balanced equation	1

Question 22 (b)

Outcomes assessed: H10, H12, H13

MARKING GUIDELINES

Criteria	Marks
• Correctly calculates moles of HCl	1

Question 22 (c)

Outcomes assessed: H10, H12, H13

MARKING GUIDELINES

Criteria	Marks
• Correctly calculates moles of NH_3	2
• Correctly calculates total moles HCl	1

Question 22 (d)

Outcomes assessed: H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
• Correctly calculates % of N to three significant figures	2
• Correctly calculates mass of N using moles of NH_3 in Part (c) OR • Correctly calculates %N to incorrect number of significant figures OR • Correctly calculates % N with incorrect mass of N	1

Question 23*Outcomes assessed: H12, H14***MARKING GUIDELINES**

Criteria	Marks
• Recognises and gives reasons for changes at 4 mins, 10 mins and 14 mins	5–6
• Recognises and gives reasons for changes at two of three times	3–4
• Recognises and gives reasons for changes at one of three times	2
• Recognises changes at three times OR • Recognises horizontal lines indicate equilibrium has been achieved OR • Gives a cause for one change	1

Question 24

Outcomes assessed: H3, H13

MARKING GUIDELINES

Criteria	Marks
• Demonstrates thorough knowledge and understanding of the principles of AAS and its environmental applications • Includes a relevant fully labelled diagram	4–5
• Demonstrates a sound knowledge and understanding of the principles of AAS and its environmental applications • Includes a relevant diagram	3
• Demonstrates a basic knowledge of the principles of AAS and its environmental applications OR • May include a basic diagram	2
• Demonstrates a limited knowledge of the principles of AAS OR its environmental applications • May include a basic diagram	1

Question 25 (a)

Outcomes assessed: H12, H14

MARKING GUIDELINES

Criteria	Marks
- Correctly calculates means - Phosphate concentrations interpolated correctly	2
- Correctly calculates means OR - Interpolation of phosphate concentrations from original absorbance	1

Question 25 (b)

Outcomes assessed: H14

MARKING GUIDELINES

Criteria	Marks
- States how THREE streams compare to recommended level - Provides TWO different sources and how these can be elevated	3
- States how THREE streams compare to recommended level - Provides any TWO sources	2
- States how THREE streams compare to recommended level OR - Provides any TWO sources OR - Compares two streams and provides one source OR - Identifies one source and cause of elevated P	1

Question 25 (c)

Outcomes assessed: H4, H14

MARKING GUIDELINES

Criteria	Marks
- States that elevated N or P causes eutrophication - States an effect of N or P	2
- States that elevated N or P causes eutrophication OR - States an effect of N or P	1

Question 26 (a)

Outcomes assessed: H10, H13

MARKING GUIDELINES

Criteria	Marks
- Gives correct oxidation and reduction half equations - Gives correct overall net ionic equation without half equations	2
- Gives correct oxidation and reduction half equations OR - Gives correct overall net ionic equation without half equations	1

Question 26 (b)

Outcomes assessed: H7, H12

MARKING GUIDELINES

Criteria	Marks
Correctly calculates $E^{\ominus}$	1

Question 26 (c)

Outcomes assessed: H13

MARKING GUIDELINES

Criteria	Marks
Correctly labels diagram	3
Correctly labels metals and ions with cathode and anode reversed	2
Correctly labels metals and ions without stating cathode and anode	1

Question 26 (d)

Outcomes assessed: H8, H11

MARKING GUIDELINES

Criteria	Marks
Names an appropriate electrolyte	1

Section II

Question 27 (a) (i)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
• States correct use	1

Question 27 (a) (ii)

Outcomes assessed: H6, H8

MARKING GUIDELINES

Criteria	Marks
- Sketches in general terms extraction, roasting and contact processes - Identifies catalyst and temperature requirement - Correctly writes TWO balanced equations or summarises chemical reactions in words	3
- Sketches ANY TWO of extraction, roasting and contact process AND EITHER - Identifies catalyst and temperature requirement OR - Writes a correctly balanced relevant equation or summarises one chemical reaction in words	2
- Outlines any process OR - Writes a relevant equation or summarises one chemical reaction in words OR - Identifies the catalyst	1

Question 27 (a) (iii)

Outcomes assessed: H4, H8

MARKING GUIDELINES

Criteria	Marks
- Identifies TWO properties of sulfuric acid - Explains how the danger is minimised for each property	2
As above for ONE property	1

Question 27 (b) (i)

Outcomes assessed: H12, H13

MARKING GUIDELINES

Criteria	Marks
• Correctly writes the equilibrium constant expression	1

Question 27 (b) (ii)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
• Correctly calculates K	3
• Incorrectly calculates K with at least TWO correct steps	2
• Identifies mole ratio	1

Question 27 (b) (iii)

Outcomes assessed: H7, H8, H14

MARKING GUIDELINES

Criteria	Marks
• Identifies effect on both K and products and/ or reactants	2
• Identifies effect on either K or products/ reactants or pressure	1

Question 27 (c)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
• Identifies a property of a named emulsion	2
• States ONE use related to properties	
• Identifies a property of a named emulsion	1

Question 27 (d) (i)

Outcomes assessed: H8, H9, H13

MARKING GUIDELINES

Criteria	Marks
- Identifies soap as having hydrophobic (non-polar) tail and a hydrophilic polar head - Shows how an emulsion forms in terms of orientation of species involved	2
ONE of the above	1

Question 27 (d) (ii)

Outcomes assessed: H9, H10, H13

MARKING GUIDELINES

Criteria	Marks
Correctly draws the structure and names the product	2
- Correctly names the product OR - Correctly draws the structure	1

Question 27 (e)

Outcomes assessed: H1, H4, H12, H13, H14

MARKING GUIDELINES

Criteria	Marks
• Demonstrates thorough knowledge and understanding of the significance of each criteria used to find a suitable location for an industrial plant to produce sodium carbonate • Refers to the flowchart • Supported by correct chemistry • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
• Demonstrates sound knowledge and understanding of the significance of each criteria to find a suitable location for an industrial plant to produce sodium carbonate • Refers to the flowchart • Communicates some scientific principles and ideas in a clear manner	4–5
• Demonstrates a basic knowledge of the significance of the criteria used to find a suitable location for an industrial plant to produce sodium carbonate • Refers to the flowchart • Communicates ideas in a basic form using general scientific terms	2–3
• Demonstrates a limited knowledge of the criteria used to find a suitable location for an industrial plant to produce sodium carbonate • Communicates simple ideas	1

Question 28 (a) (i)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• States aluminum is used in drink cans in Australia	1

Question 28 (a) (ii)

Outcomes assessed: H6, H8

MARKING GUIDELINES

Criteria	Marks
- Shows similarities or differences between the use and effectiveness of the three cans - Relates use to coating	3
- Clearly explains the use and effectiveness for one can OR - Outlines the use and effectiveness for three cans	2
- Identifies a coating used in tins OR - States the coating acts as a barrier OR - States the barrier is only effective whilst intact OR - Identifies tin as a less reactive metal than iron	1

Question 28 (b)

Outcomes assessed: H4, H6, H8

MARKING GUIDELINES

Criteria	Marks
- States galvanised steel is zinc coated - Identifies Zn as a passivating metal providing protective barrier - States that zinc can provide electrons to iron since it is more reactive	2
- States zinc is passivating metal OR - Zinc is more reactive than iron OR - States galvanised steel is zinc coated	1

Question 28 (c) (i)

Outcomes assessed: H10, H13

MARKING GUIDELINES

Criteria	Marks
- Shows the cause of rusting by correctly writing the equation for the reduction of water - Correctly writing the equation for the oxidation of iron	2
- Correctly writes ONE relevant equation OR - States that iron is oxidised as the oxygen is reduced	1

Question 28 (c) (ii)

Outcomes assessed: H8, H12, H13

MARKING GUIDELINES

Criteria	Marks
- Relates reactivity of the three metals Al, Fe and Cu to their electrode potentials - States consequences of galvanic coupling in terms of corrosion for both types of guttering - Justifies the action taken	4
- Ranks the three metals in terms of reactivity - States consequences of galvanic coupling in terms of corrosion for one type of guttering - Justifies the action taken	3
TWO of below	2
- Justifies the action taken OR - Ranks the three metals in terms of reactivity OR - States that the more active metal will corrode when two different metals are paired OR - Gives standard electrode potentials for the three metals	1

Question 28 (d) (i)

Outcomes assessed: H6, H8

MARKING GUIDELINES

Criteria	Marks
- Shows how the electrolytic process conserves this artefact by: - Stating that electrolysis is used to remove (embedded) Cl^- - Identifying artefact is cathode (and stainless steel is anode) - Outlining chemistry of process	3
TWO of the above	2
- States Cl^- is removed by electrolysis OR - Identifies artefact is cathode OR - Outlines chemistry of process	1

Question 28 (d) (ii)

Outcomes assessed: H6, H8, H13

MARKING GUIDELINES

Criteria	Marks
- Draws a fully labelled diagram - Describes the process	3
- Draws a fully labelled diagram OR - Describes the process	2
- Draws a diagram OR - Identifies the cathode or anode OR - Outlines the process	1

Question 28 (e)

Outcomes assessed: H1, H2, H3, H7, H8

MARKING GUIDELINES

Criteria	Marks
• Demonstrates thorough knowledge and understanding of how the work of each scientist has led to a better understanding of electron transfer reactions • Refers to the flowchart • Supports by the use of an equation • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
• Demonstrates sound knowledge and understanding of how the work of each scientist has led to a better understanding of electron transfer reactions • Refers to the flowchart • Communicates some scientific principles and ideas in a clear manner	4–5
• Demonstrates a basic knowledge of how the work of each scientist has led to a better understanding of electron transfer reactions • Refers to the flowchart • Communicates ideas in a basic form using general scientific terms	2–3
• Demonstrates a limited knowledge of the work of the scientist(s) or an understanding of an electron transfer reactions • Communicates simple ideas	1

Question 29 (a) (i)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Correctly identifies a factor	1

Question 29 (a) (ii)

Outcomes assessed: H6, H8, H9

MARKING GUIDELINES

Criteria	Marks
- Relates denaturation to disruption of the chemical bonding forces responsible for secondary and tertiary structures of proteins - Identifies the particular bonding forces disrupted	2
- Relates denaturation to disruption of the chemical bonding forces responsible for secondary and tertiary structures of proteins OR - Identifies TWO bonding forces disrupted	1

Question 29 (b)

Outcomes assessed: H9, H14

MARKING GUIDELINES

Criteria	Marks
- Identifies and related colour change to enzyme activity between pH 4.0–7.0 - Relates dark red at pH 6.0–7.0 to extent of oxidation - Concludes that experiment confirms enzymes act optimally within narrow pH ranges	3
TWO of the above	2
- States most enzymes act optimally within narrow pH ranges OR - Relates dark red to optimal pH for enzyme activity OR - Relates experimental colour change to enzyme activity	1

Question 29 (c) (i)

Outcomes assessed: H9, H10, H12

MARKING GUIDELINES

Criteria	Marks
- Gives correctly balanced equation - Gives correct ΔH	2
ONE of the above	1

Question 29 (c) (ii)

Outcomes assessed: H9, H12, H13

MARKING GUIDELINES

Criteria	Marks
- Calculates correct answer OR - Correctly calculates answer from incorrect kJ	2
- Correct method with incorrect formula mass OR - Gives correct formula mass OR - Calculates correct answer without correct units	1

Question 29 (c) (iii)

Outcomes assessed: H12, H13

MARKING GUIDELINES

Criteria	Marks
- Calculates correct answer OR - Calculates correct answer from incorrect kJ	2
- Uses correct steps with incorrect answer OR - Uses one step correctly OR - Calculates correct answer without correct units	1

Question 29 (d) (i)

Outcomes assessed: H6, H9

MARKING GUIDELINES

Criteria	Marks
- Describes structure of glycerol or gives diagram - Relates solubility due to polar nature - Relates viscosity due to H-bonds	3
TWO of the above	2
ONE of the above	1

Question 29 (d) (ii)

Outcomes assessed: H6, H9

MARKING GUIDELINES

Criteria	Marks
- States fatty acids have a hydrophilic end from carboxylic group - States fatty acids have a hydrophobic end from hydrocarbon group - States long chain fatty acids are insoluble in water - Identifies triacylglycerols are esters with no O–H group, therefore non-polar not soluble in H₂O	3
Identifies ONE property of both fatty acids and triacylglycerols related to solubility	2
Identifies ONE property of either fatty acids or triacylglycerols related to solubility	1

Question 29 (e)

Outcomes assessed: H3, H8, H9, H13, H14

MARKING GUIDELINES

Criteria	Marks
• Demonstrates thorough knowledge and understanding of the biochemical reactions involved in muscle cell respiration that have led to improved training programs • Refers to flowchart • Includes relevant chemical equations • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
• Demonstrates sound knowledge and understanding of biochemical reactions involved in muscle cell respiration that have led to improved training programs • Refers to flowchart • Includes relevant chemical equations • Communicates some scientific principles and ideas in a clear manner	4–5
• Demonstrates a basic knowledge of biochemical reactions involved in muscle cell respiration • Refers to flowchart • Communicates ideas in a basic form using general scientific terms	2–3
• Demonstrates a limited knowledge of biochemical reactions involved in muscle cell respiration • Communicates simple ideas	1

Question 30 (a) (i)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Gives a correct ligand	1

Question 30 (a) (ii)

Outcomes assessed: H6, H9

MARKING GUIDELINES

Criteria	Marks
- Defines term ‘polydentate’ - Identifies donor atoms in EDTA	2
ONE of the above	1

Question 30 (a) (iii)

Outcomes assessed: H8, H9

MARKING GUIDELINES

Criteria	Marks
- Identifies bonds as coordinate covalent bonds - States that pairs of electrons come from the donor atoms - States that these electron pairs interact with d-orbitals of transition metals	3
TWO of the above	2
ONE of the above	1

Question 30 (b) (i)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
Gives correct maximum number of electrons	1

Question 30 (b) (ii)

Outcomes assessed: H6, H13

MARKING GUIDELINES

Criteria	Marks
Gives all THREE electron configurations correctly	2
Gives correct electron configuration for excited Ca atom or Ca^+ ion	1

Question 30 (c)

Outcomes assessed: H6, H12

MARKING GUIDELINES

Criteria	Marks
- Provides explanation for increase in IE across a period - Gives explanation on each of the drops from 2–3, 10–11, 18–19 EITHER - Explains an irregularity within a period OR - Explains the difference between 2nd period (2–10) and 3rd period (11–18)	3
TWO of the above	2
ONE of the above	1

Question 30 (d) (i)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
- States that atoms can exist in an excited electronic state - States that light is emitted at particular wavelengths as the electrons fall back to the ground state, to a lower energy level - States these observations were the basis of Planck's Quantum Theory - States that particular wavelengths correspond to specific changes in energy levels which in turn correspond to particular colours of the spectrum	3
TWO of the above	2
ONE of the above	1

Question 30 (d) (ii)

Outcomes assessed: H6, H11

MARKING GUIDELINES

Criteria	Marks
- Provides detailed explanation of appropriate method for analysis - Gives a reason why validity might be compromised	3
- Outlines a valid method - Gives ONE reason why the validity might be compromised	2
- Gives ONE valid method OR - Gives ONE reason why the validity might be compromised	1

Question 30 (e)

Outcomes assessed: H1, H6, H9, H13, H14

MARKING GUIDELINES

Criteria	Marks
• Demonstrates thorough knowledge and understanding of the use of pigments (with atomic number in the range of 22–30) and the origin of the colour at the atomic level • Refers to the flowchart • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
• Demonstrates sound knowledge and understanding of the use of pigments (with atomic number in the range of 22–30) and the origin of the colour at the atomic level • Refers to the flowchart • Communicates some scientific principles and ideas in a clear manner	4–5
• Demonstrates a basic knowledge of the use of pigments and the origin of the pigment colour • Refers to the flowchart • Communicates ideas in a basic form using general scientific terms	2–3
• Demonstrates a limited knowledge of the use of pigments or the origin of the pigment colour • Communicates simple ideas	1

Question 31 (a) (i)

Outcomes assessed: H6, H9, H12

MARKING GUIDELINES

Criteria	Marks
Correctly names bond	1

Question 31 (a) (ii)

Outcomes assessed: H6, H9

MARKING GUIDELINES

Criteria	Marks
Identifies differences between 1°, 2°, 3° structures clearly	3
- Identifies differences between TWO levels of structure OR - Identifies 1° structure and indicates possible 2°, 3° structures being higher level of complexity overlaid	2
Identifies ONE level of structure	1

Question 31 (a) (iii)

Outcomes assessed: H8, H9

MARKING GUIDELINES

Criteria	Marks
- Outlines electrophoresis principle - States proteins in samples are subject to electrophoresis to allow comparison between known and unknown proteins	2
- Outlines principle OR - States that proteins can be compared via electrophoresis	1

Question 31 (b) (i)

Outcomes assessed: H12

MARKING GUIDELINES

Criteria	Marks
Identifies the ship responsible for spill	1

Question 31 (b) (ii)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
- States each peak represents a separate component - Identifies that the signal intensity is proportional to the amount of component	2
- States each peak represents a separate component OR - Identifies that the signal intensity is proportional to the amount of component	1

Question 31 (c)

Outcomes assessed: H3, H6

MARKING GUIDELINES

Criteria	Marks
Provides features of THREE components of either GLC or HPLC systems that allow for smaller sample analysis	3
Provides features of TWO components of either GLC or HPLC systems that allow for smaller sample analysis	2
Provides features of ONE component of either GLC or HPLC system that allows for smaller sample analysis	1

Question 31 (d) (i)

Outcomes assessed: H9, H11

MARKING GUIDELINES

Criteria	Marks
- Clearly details procedures in a logical order - Compares chromatogram for identification	3
- Clearly outlines method and comparison of chromatogram for identification OR - Clearly details procedures in a logical order	2
Outlines method	1

Question 31 (d) (ii)

Outcomes assessed: H8, H9

MARKING GUIDELINES

Criteria	Marks
Provides features of the tests which conclusively identify each of 3 species unambiguously	3
- Outlines more than ONE test unambiguously OR - Outlines tests which confirm each of the 3 species	2
Outlines a test	1

Question 31 (e)

Outcomes assessed: H3, H9, H14

MARKING GUIDELINES

Criteria	Marks
• Demonstrates thorough knowledge and understanding of the processes forensic scientists follow from sample collection to presenting results • Refers to flowchart • Demonstrates coherence and logical progression and includes correct use of scientific principles and ideas	6–7
• Demonstrates sound knowledge and understanding of the processes forensic scientists follow from sample collection to presenting results • Refers to flowchart • Communicates some scientific principles and ideas in a clear manner	4–5
• Demonstrates a basic knowledge of the processes forensic scientists follow • Refers to the flowchart • Communicates ideas in a basic form using general scientific terms	2–3
• Demonstrates a limited knowledge of some of the processes forensic scientists follow • Communicates simple ideas	1

Chemistry

2009 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
Section I			
Part A			
1	1	9.2.5.2.1	H6
2	1	9.3.2.2.1, 9.3.2.2.5	H8
3	1	9.3.1.2.1	H8
4	1	9.4.3.3.1	H6
5	1	9.2.3.3.6, 11.2a	H11
6	1	9.2.1.3.2	H9
7	1	9.3.4.2.3, 9.3.4.2.5, 13.1d	H12, H13
8	1	9.4.3.3.1, 9.4.3.2.1, 11.3a	H8, H11
9	1	9.2.4.2.3, 12.3c, 12.4b	H6, H10, H12, H14
10	1	9.3.5.2.4, 9.2.1.3.1	H9
11	1	9.4.5.2.3, 9.4.5.3.3, 12.3c	H8, H12
12	1	9.4.4.2.9	H9, H13
13	1	9.2.3.2.6, 9.2.3.3.5, 12.4b	H9, H10, H12
14	1	9.3.4.3.3, 12.4b	H10, H12
15	1	9.4.5.2.1, 9.3.2.2.9, 12.4b	H10, H12
Section I			
Part B			
16	3	9.3.5.3.1, 9.3.5.2.4, 9.3.5.2.5, 9.3.5.2.6, 12.1a–d	H12
17	4	9.2.3.2.3, 13.1e, 14.1g,h	H2, H4, H13, H14
18	5	9.3.2.2.6, 9.3.2.2.7, 9.3.2.2.8, 9.3.2.2.10, 9.3.2.3.2, 9.4.1.2.3, 9.4.4.2.2	H4, H6, H8
19	6	9.2.1, 13.1d, 14.3b	H9, H13, H14
20 (a)	3	9.2.3.2.7, 9.2.3.3.6, 12.4b	H12
20 (b)	1	9.2.3.3.6, 11.2c,d,e, 11.3a	H11
21 (a)	3	9.3.4.3.3, 9.3.3.2.4, 12.3c, 14.1a–f	H12, H14
21 (b)	1	9.3.4.2.4, 9.3.4.3.3, 9.3.3.3.4	H8, H10
21 (c)	1	9.3.3.3.7, 12.3c, 12.4b, 13.1d	H12, H13
21 (d)	1	9.3.1.2.2, 9.3.1.3.2	H10
22 (a)	2	9.2.1.2.1, 9.3.4.3.3, 13.1d	H10, H13
22 (b)	1	9.3.4.3.3, 12.4b, 13.1d	H10, H12, H13
22 (c)	2	9.3.4.3.3, 12.4b, 13.1d	H10, H12, H13
22 (d)	2	9.4.3.1, 9.4.3.3.3, 12.4b, 13.1d, 14.1h	H12, H13, H14
23	6	9.3.2.2.3, 9.3.2.2.4, 9.4.2.1, 12.3c, 14.1a–h	H12, H14
24	5	9.4.3.2.2, 9.4.3.3.5, 9.4.5.3.2, 13.1e	H3, H13
25 (a)	2	9.4.3.3.1, 9.4.3.2.1, 12.3c	H12

Question	Marks	Content	Syllabus outcomes
25 (b)	3	9.4.5.2.2, 9.4.5.3.3, 14.1g	H14
25 (c)	2	9.4.5.2.1, 9.4.5.2.2, 9.4.3.3.2, 14.3d	H4, H14
26 (a)	2	9.2.4.2.1, 9.2.4.3.3, 13.1d	H10, H13
26 (b)	1	9.2.4.3.4, 12.4b	H7, H12
26 (c)	3	9.2.4.2.5, 9.2.4.2.6, 13.1e	H13
26 (d)	1	9.2.4.2.6, 9.4.3.3.1, 11.3a	H8, H11

Section II

Question 27 Industrial Chemistry

(a) (i)	1	9.5.3.2.1	H8
(a) (ii)	3	9.5.3.2.2, 9.5.3.2.3, 9.5.3.3.1	H6, H8
(a) (iii)	2	9.5.3.3.3	H4, H8
(b) (i)	1	9.5.2.2.2, 12.4b, 13.1a	H12, H13
(b) (ii)	3	9.5.2.3.3, 12.4b	H12
(b) (iii)	2	9.5.2.2.1, 9.5.2.2.3, 14.3d	H7, H8, H14
(c)	2	9.5.5.3.3	H9
(d) (i)	2	9.5.5.2.3, 9.5.5.2.4, 13.1e	H8, H9, H13
(d) (ii)	2	9.5.5.2.1, 13.1d	H9, H10, H13
(e)	7	9.5.6.2.1, 9.5.6.2.3, 9.5.6.2.4, 9.5.6.2.2, 9.5.6.3.3, 12.3c, 13.1d, 14.3b	H1, H4, H12, H13, H14

Section II

Question 28 Shipwrecks, Corrosion and Conservation

(a) (i)	1	9.6.2.2.1	H6
(a) (ii)	3	9.6.4.3.3	H6, H8
(b)	2	9.6.4.2.3/4, 9.6.4.3.3/4	H4, H6, H8
(c) (i)	2	9.6.2.2.4, 9.6.4.3.2, 13.1d	H10, H13
(c) (ii)	4	9.6.4.2.2, 12.4b, 13.1d	H8, H12, H13
(d) (i)	3	9.6.3.2.1, 9.6.7.2.1/3/4, 9.6.7.3.1	H6, H8
(d) (ii)	3	9.6.3.2.1/2, 13.1e	H6, H8, H13
(e)	7	9.6.1.2.2/3/4, 9.6.1.3.1, 14.3b	H1, H2, H3, H7, H8, H14

Section II

Question 29 The Biochemistry of movement

(a) (i)	1	9.7.4.2.6, 9.7.4.3.2	H9
(a) (ii)	2	9.7.4.2.5, 9.7.4.2.6	H6, H8, H9
(b)	3	9.7.4.3.2, 9.7.4.3.3, 14.1c	H9, H14
(c) (i)	2	9.7.7.9.7.8, 9.7.9, 9.7.7.1, 9.7.8.1, 9.7.8.3.1, 12.3c, 12.4b	H9, H10, H12
(c) (ii)	2	12.3c, 12.4b, 13.1d	H9, H12, H13
(c) (iii)	2	12.3c, 12.4b, 13.1d	H12, H13
(d) (i)	3	9.7.3.3.2	H6, H9
(d) (ii)	3	9.7.3.2.1, 9.7.3.3.1, 9.7.3.2.2, 9.7.3.2.5	H6, H9
(e)	7	9.7.2.2.2/3, 9.7.2.3.1, 9.7.5.1, 9.7.10.1, 13.1d, 14.3b	H3, H8, H9, H13, H14

Question	Marks	Content	Syllabus outcomes
Section II			
Question 30 The Chemistry of Art			
(a) (i)	1	9.8.5.2.5	H9
(a) (ii)	2	9.8.5.2.4, 9.8.5.2.6	H6, H9
(a) (iii)	3	9.8.5.2.3, 9.8.5.3.1, 9.8.5.2.6	H8, H9
(b) (i)	1	9.8.3.2.1, 9.8.3.2.2	H6
(b) (ii)	2	9.8.3.2.4, 9.8.3.2.5, 13.1b	H6, H13
(c)	3	9.8.3.2.6, 9.8.3.2.8, 9.8.3.3.1, 12.3c	H6, H12
(d) (i)	3	9.8.2.2.3	H6
(d) (ii)	3	9.8.2.3.1, 11.2c	H6, H11
(e)	7	9.8.1.1, 9.8.4.2.3, 9.8.4.2.4, 9.8.5.1, 13.1d, 14.3b	H1, H6, H9, H13, H14
Section II			
Question 31 Forensic Chemistry			
(a) (i)	1	9.9.3.2.4, 12.3c	H6, H9, H12
(a) (ii)	3	9.9.3.2.1, 9.9.3.3.1	H6, H9
(a) (iii)	2	9.9.3.2.5, 9.9.3.2.6	H8, H9
(b) (i)	1	9.9.5.2.1, 9.9.5.2.2, 12.3c	H12
(b) (ii)	2	9.9.5.2.2	H6
(c)	3	9.9.5.2.2	H3, H6
(d) (i)	3	9.9.3.3.4, 11.2c	H9, H11
(d) (ii)	3	9.9.1.2.3, 9.9.1.3.3	H8, H9
(e)	7	9.9.1.2.1, 9.9.2.1, 9.9.3.1, 9.9.4.1, 9.9.5.1, 14.3b	H3, H9, H14